Ismail Ashercook's CV

My Portfolio

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Personal Statement

A third-year Software Engineering student at Heriot-Watt University, Ismail Ashercook is passionate about learning and developing reliable software for real-time and embedded systems. He has engineered systems including a health care website, a student portal website, a portfolio website, a library management system website. He is seeking to apply his software engineering skills, both theoretical and practical, to industrial challenges while advancing his background through professional mentorship.

Skills

Technical Skills:

- **Programming Languages:** Java (3 years), C# (1 year), Python (3 years), ARM Assembly (1 year), HTML (3 years), CSS (3 years), JavaScript (2 years), Node.js (1 year), PHP (2 years), SQL (3 years).
- **Software Engineering & Design:** Object-Oriented Programming (OOP), System modelling and documentation using UML (Unified Modelling Language), Clean Code Principles, Agile Methodologies.
- **Systems & Tools:** Windows, Linux, MySQL, MongoDB, Microsoft Office, Figma, JustInMind, Visual Studio, Eclipse IDE, pgAdmin, GNS3, Ubuntu, React.
- **Core Competencies:** Analytical problem-solving, ability to analyse time/space complexity, debugging, and systematic testing. Strong collaborator and team leader.

Projects

Student Portal Website (University Group Project)

Purpose & Context: Designed and delivered a full-stack web application to streamline academic interactions for students and teachers, providing role-based access to schedules, resources, and assignments.
Challenges & Approach: Collaborated within an **Agile (Scrum)** team of peers to develop a **responsive front end** using **HTML, CSS, and JavaScript**, supported by a **Node.js backend with MongoDB**. Implemented **secure user authentication** and **role-based dashboards (student/teacher)**, focusing on **scalability** and **usability** informed by peer feedback.

Health Care System (University Group Project)

Purpose & Context: Developed a **full-stack** healthcare web application to enable patients to monitor health metrics including heart rate, blood pressure, and activity data, providing a centralized platform for personal health tracking and administrative oversight.

Challenges & Approach: Built **responsive frontend with React** (functional components, hooks, state management) and **backend REST API** with **Node.js/Express**. Implemented secure **JWT-based authentication** with login/signup functionality, role-based access control, and an admin dashboard to view registered user details. Designed a personalized metrics dashboard fetching real-time and historical health data from a **MySQL database**. Integrated data visualization for heart rate trends and other biometrics, ensuring secure **data handling** and **responsive design** across devices.

Library Management System (University Group Project)

Purpose & Context: Led the development of a digital system to automate library operations—tracking loans, returns, and book availability—in a scalable, user-friendly environment.
Challenges & Approach: Directed a **5-person Agile team**, creating **UML diagrams (class,**

sequence) to model system architecture. Engineered a **high-performance Java backend** with **data validation** and **systematic testing** to ensure **data integrity**. Presented the final solution to lecturers, demonstrating **clear communication of technical decisions**.

AI Sudoku Solver (Personal Project)

Purpose & Context: Designed and implemented an intelligent Sudoku solver using **Constraint Satisfaction Problem (CSP)** techniques to efficiently solve any valid 9×9 puzzle, demonstrating core AI search and constraint propagation.

Challenges & Approach: Built a **backtracking** solver with **forward checking** and **MRV heuristic** in **Java**, significantly reducing search space and solve time. Formally modelled Sudoku as a **CSP** with 81 variables, domain {1..9}, and row/column/subgrid **constraints**. Compared theoretical performance against **A* search** analysing complexity and **heuristic** viability. Validated 100% accuracy with **JUnit** tests across 30+ puzzles. Designed modular architecture separating solver, model, and **GUI**.

Experience

Computer Science Tutor | September 2023 – Present
FindTutors, two students

- Providing one-to-one tuition in Python, SQL, and web technologies, improving student problem-solving and exam readiness.
- Tailoring explanations to individual learning styles, demonstrating patience and clear communication.

Work Insight Week | November 2022
CAPCO – Financial Technology Consultancy

- Gained exposure to Agile workflows, client engagement, and the dynamics of cross-functional teamwork in developing technology-driven solutions.

Maths Prefect (Volunteer) | June 2022 – June 2023
Craigmount High School`

- Supported teachers and mentored peers, developing leadership and collaborative skills.

Education

2023-2028 MEng Software Engineering, Heriot-Watt University, Edinburgh, Second-year average 74% (across all second-year modules).

Relevant modules: Software Development, Discrete Mathematics, Introduction to Computer Systems, Programming Languages, Data Structures and Algorithms, Hardware-Software Interface, Introduction to Software Engineering, Artificial Intelligence and Intelligent Agents.

Awards

- Bebras Computing Challenge – recognised for strong performance in a national computational thinking and problem-solving competition.
- Advanced Higher Award in Computing Science – school achievement award for excellence.
- Higher Award in Computing Science – school achievement award for excellence.
- Saltire Award – Scottish Government recognition for sustained community service and leadership.

Hobbies and Interests

- Aikido martial arts: learning martial arts skills and keeping active.
- Recreational Swimming: maintaining fitness and keeping active.
- Member of the Gaming Society: Participating in gaming tournaments and friendly competitions.
- Member of the Coding Society: Regularly engage in hackathons and LeetCode challenges with peers, while offering debugging assistance and constructive code reviews to members.